

3. **Course Title: Construction Materials**

Course Code.: CE126

Credit: 3

Number of hours per week: 2

Nature of the Course: Theory

Total hours: 45

Year/Semester: First/Second

Level: Bachelor of Engineering (Civil)

1. **Course Introduction:**

The course is aimed to preparing students to understand the fundamentals of construction materials. This course provides an introductory overview of the various materials used in construction. Resulting from this course, students will gain a comparative knowledge of material properties and possible applications in construction and architecture.

2. **Course Objectives:**

- At the end of this course the student should be able to understand the fundamentals of construction materials used in construction
- Introduce students to the science and technology of construction materials.
- Review important material properties
- Teach students how to select appropriate construction
- Teach technologies of basic construction materials, such as steel, concrete, asphalt, wood, and polymers and composite materials.

3. **Specific Objectives and Contents:**

Specific Objectives	Contents	
•	UNIT 1.	INTRODUCTION (3 Hours)
• To know the classification, qualities and uses of commonly used building stones • To understand the process and importance of dressing of stones. • Understand the causes of decay of stones, their preservation and tests of stones.	UNIT 2.	BUILDING STONES (5 Hours)
	2.1	Common building stones and their uses
	2.2	Quality of good building stones
	2.3	Test for stones
	2.4	Deterioration of stones
	2.5	Preservation of stones
• Understand the importance of clay as a building construction material. • To understand the process of manufacturing good quality bricks and tiles and their uses.	2.6	Dressing of stones
	UNIT 3.	BRICKS & TILES (5 Hours)
	3.1	Classification of bricks
	3.2	Manufacture of bricks
	3.3	Good Quality of bricks
	3.4	Tiles and their types
• Understand the importance of timber products as a building construction material and their uses.	3.5	Quality of tiles & their uses
	UNIT 4.	TIMBER (5 Hours)
	4.1	Varieties & uses
	4.2	Defects in Timber
• To understand the process and importance of seasoning timber. • Understand the causes of decay of timber, their preservation and tests of timber.	4.3	Tests for good Timber
	4.4	Deterioration and Preservation of Timber
	4.5	Seasoning of timber
	4.6	Products of timber & their uses

• Understand the type, properties and uses of lime and cement. • Understand the tests for cement. • To know the manufacturing process of an ordinary cement. • Understand the type, properties and uses of lime and cement mortar.	UNIT 5. LIME & CEMENT (7 Hours) 5.1 Type, Properties and uses of lime 5.2 Type, Properties and uses of cement 5.3 Constituents of Cement 5.4 Manufacture of an Ordinary Cement 5.5 Chemical Composition and Hydration of Ordinary Cement 5.6 Laboratory and standard tests for cement 5.7 Admixtures 5.8 Lime & cement mortar
• To know the properties and ingredients of concrete cement. • To understand the importance of good quality cement concrete. • To get brief idea about pre-cast concrete and R.C.C. work. • To understand the properties and handing of fresh concrete to gain the maximum strength.	UNIT 6. Cement Concrete (10 Hours) 6.1 Properties and ingredients of cement concrete ▪ Normal consistency, setting time, soundness. ▪ Compression strength of cement ▪ Grades of cement ▪ Quality of mixing water. ▪ Grading of aggregates and importance of size, shape and texture. ▪ Fine aggregate ▪ Coarse aggregate ▪ Water cement ratio 6.2 Materials used in R.C.C. work 6.3 Water proofing cement concrete 6.4 Pre-cast concrete 6.5 Fresh Concrete ▪ Workability – factors affecting workability, ▪ Measurement of workability ▪ Compaction factor and vee-bee consistometer tests. ▪ Segregation and bleeding. ▪ Process of manufacture of concrete: Batching. ▪ Mixing. ▪ Transporting ▪ Placing ▪ Compaction ▪ Curing ▪ Chemical admixtures- plasticizers, accelerators, retarders and
	- entraining agents. ▪ Mineral admixtures – fly ash. ▪ Silica fumes and rice husk ash. 6.6 Joints in concrete structure 6.7 Quality control of concrete

- Understand the importance of metal products as a building construction material and their uses. - Understand the type, properties and uses of metal and alloy as an engineering material. - To understand the process of corrosion and method of prevention.	UNIT 7. Metals and Alloys (5 Hours) 7.1 General Introduction 7.2 Type, properties and uses of iron 7.3 Type, properties and uses of steel 7.4 Nonferrous metals 7.5 Steel alloys 7.6 Corrosion - Causes of corrosion and factor influencing corrosion - Effect of corrosion-Ferrous and nonferrous metals - Prevention of corrosion
Understand the importance of other miscellaneous materials and their products as a building construction material and their uses.	UNIT 8. Properties & uses : Other building materials (5 Hours) 8.1 Reinforcing steel, structural steel 8.2 Cast Iron, Plain carbon steel 8.3 Glasses 8.4 Electrical, Thermal & Sound Insulating Materials 8.5 Paints, Varnish & Enamels 8.6 Plastics 8.7 Rubber 8.8 Gypsum Products 8.9 Asphalt, Bitumen and Tar

Evaluation System

Undergraduate Programs				
External Evaluation	Marks	Internal Evaluation	Weight age	Marks
End semester examination (Details are given in the separate table at the end)	60	Assignments	10%	40
		Quizzes	10%	
		Attendance	10%	
		Presentation	10%	
		Mid-Term & Pre-board exam	50%	
		Group work	10%	
Total External	60	Total Internal	100%	40

External evaluation

End semester examination

It is a written examination at the end of the semester. The questions will be asked covering all the units of the course. The question model, full marks, time and others will be as per the following grid.

Full Marks: 100, Pass Marks: 45, Time: 3 Hrs

Nature of question	Total questions to be asked	Total questions to be answered	Total marks	Weightage	External exam marks
Group A: multiple choice*	20	20	$20 \times 1 = 20$	20%	12
Group B: Short answer type questions	8	6	$6 \times 8 = 48$	40%	24
Group C: Long answer type question	3	2	$2 \times 16 = 32$	40%	24
			100	100%	60

Each student must secure at least 50% marks in internal evaluation in order to appear in the end semester examination. Failed student will not be eligible to appear in the end semester examinations.

Internal evaluation

Assignment: Each student must submit the assignment individually. The stipulated time for submission of the assignment will be seriously taken.

Quizzes: Unannounced and announced quizzes/tests will be taken by the respective subject teachers. Such quizzes/tests will be conducted twice per semester. The students will be evaluated accordingly.

Attendance in class: Students should regularly attend and participate in class discussion. Eighty percent class attendance is mandatory for the students to enable them to appear in the end semester examination. Below 80% attendance in the class will signify NOT QUALIFIED (NQ) to attend the end semester examination.

Presentation: Students will be divided into groups and each group will be provided with a topic for presentation. It will be evaluated individually as well as group-wise. Individual students have to make presentations on the given topics.

Mid-term & Pre-board examination: These are written examination and the questions will be asked covering all the topics in the session of the course.

Discussion and participation: Students will be evaluated on the basis of their active participation in the classroom discussions.

Instructional Techniques: All topics are discussed with emphasis on real-world application. List of instructional techniques is as follows:

- Lecture and Discussion
- Group work and Individual work
- Assignments
- Presentation by Students
- Term Paper writing
- Case study
- Quizzes
- Guest Lecture

Students are advised to attend all the classes and complete all the assignments within the specified time period. If a student does not attend the class(es), it is his/her sole responsibility to cover the topic(s) taught during that period.

Prescribed Text:

1. "Engineering Materials", Rangawala P.C. Charter Publishing House, Anand, India.

References:

1. "Fundamental of Engineering Materials", Peter A. Thomson and Vito J. Colangelo, Prentice Hall College Div, 1985.
2. "Engineering Materials", Sushil Kumar, Standard Publication and Distributors, New Delhi.
3. "Engineering Material", R.K. Rajput, S. Chand & Company Ltd, 2004